BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Protocols and Layers

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does a wireless access point do?

2. Why can a mesh survive some broken links?

A) It has only one path
B) Every device is a server
C) It has other paths
D) Every link is wireless

3. "Fibre carries electrical signals through copper wire." Correct this. Explain why.

4. Desktops and moving tablets need network links. "Wi-Fi needs a cable from each tablet to the switch." Correct this. Which links suit these devices?

Today's objectives

Explain the core ideas of network protocols and layers using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

IP address
Identifies devices/interfaces for routing across IP networks.

MAC address
Identifies a network interface at the data-link level for local communication.

DNS
Resolves a domain name to an IP address.

Protocols and layers
Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

IP address	Identifies devices/interfaces for routing across IP networks.
MAC address	Identifies a network interface at the data-link level for local communication.
DNS	Resolves a domain name to an IP address.
Protocols and layers	Agreed rules such as HTTP(S), TCP/IP and SMTP support interoperable communication.

I Do - One worked example

A pupil enters a website domain name into a browser. Explain DNS and packet delivery.

1. The browser needs the server's IP address.
2. DNS provides the mapping from domain name to IP.
3. Packets can be routed to the destination across networks.
4. The browser and server use suitable protocols, e.g. HTTPS for encrypted web transfer.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Explain how a school laptop sends an email.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that recognises a secure web protocol.

```
protocol = "HTTPS"
if ... then
print("Secure web")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the purpose of DNS. [2 marks]

Q3. Distinguish IP and MAC addresses. [4 marks]

Q4. Name a suitable protocol for sending email. [1 mark]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. DNS translates a domain name into an IP address. ☐ T ☐ F
2. Identifies devices/interfaces for routing across IP networks. ☐ T ☐ F
3. Identifies a network interface at the data-link level for local communication. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It enables wireless devices to connect to the network. ☐

Q2 C - It has other paths ☐

Q3 Fibre uses light. It supports fast data transfer over long distances and resists electromagnetic interference. Installation can cost more. ☐

Q4 Ethernet uses cables. Wi-Fi uses radio, so tablets can move around, but interference can affect it. Choose links to suit the distance and data rate needed. ☐

Code First - Model

```
protocol = "HTTPS"
if protocol == "HTTPS" then
print("Secure web")
endif
```

Output: Secure web

We Do - Check the model

1. Avoid confusing SMTP with IMAP/POP reception. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Resolves domain names to IP addresses. ☐

Q3 IP used for logical addressing/routing across networks; MAC identifies a network interface for local data-link forwarding. ☐

Q4 SMTP. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: